

Palak Gupta

9990072250 | palakgupta425@gmail.com | [linkedin.com/in/palakgupta5412](https://www.linkedin.com/in/palakgupta5412) | github.com/palakgupta5412

SUMMARY

Information Technology undergraduate with expertise in both **Artificial Intelligence** and **MERN-Stack Web Development**. Skilled in building intelligent AI-powered applications using LLMs, OCR pipelines, RAG systems, and scalable MERN-stack architectures. Experienced in integrating Generative AI APIs into modern web applications to solve real-world problems.

EDUCATION

Maharaja Surajmal Institute of Technology
Bachelor of Technology in Information Technology

New Delhi, India
Sep. 2023 – June 2027

EXPERIENCE

AI & Software Engineering Intern
Orbit Pro / Remote

April 2026 – Present
New Delhi, India

- Architected and developed Agentic AI pipelines using Python, LangChain, multimodal LLMs, and RAG to automate healthcare document intelligence, structured data extraction, and medical content generation.
- Built Intelligent Document Processing (IDP) workflows to parse 6,000+ pages of medical PDFs, extracting previous-year questions, diagrams, and metadata into validated JSON schemas powering the GO-MED question bank.
- Designed multimodal Vision-Language AI and prompt-chaining workflows to generate clinically accurate, image-based MCQs with structured outputs, contributing to the production database of GO-MED (4K+ active users).
- Contributed to full-stack product development, implementing frontend enhancements for GO-MED and maintaining feature updates for Dr. Jasmine Kalsi's website while collaborating on scalable AI product development.

PROJECTS

CTRL : AI Productivity Engine | *React, Tailwind, Groq API* | [GitHub](#) | [Live Demo](#)

- Engineered a **Manifest V3** Chrome Extension acting as a strict productivity firewall, utilizing **React** and **Tailwind CSS** to deploy aggressive, physics-based UI interventions.
- Integrated the blazing-fast **Groq API (LLaMA-3.1)** to perform asynchronous, real-time **DOM text scraping** and contextual intent analysis, intercepting distracting content within milliseconds.
- Architected a memory-safe, **stateless heartbeat engine** via the **chrome.alarms API** to bypass background script limitations, powering an unbreakable "Deep Work Vault" to strictly enforce focus sessions.

Better With Books | *React, Node.js, Gemini AI, GSAP* | [GitHub](#) | [Live Demo](#)

- Engineered a seamless, interactive virtual library platform utilizing **React** and **GSAP** to deliver a cinematic, physics-based "digital bookshelf" user interface.
- Integrated the **Google Books API** with a **Gemini AI** recommendation engine, parsing user preferences to dynamically generate and serve highly personalized reading lists.
- Optimized frontend rendering and state management, ensuring smooth 60fps animations and reducing visual load times for a highly engaging user experience.

Planify | *MERN Stack, Cloudinary, REST APIs* | [GitHub](#) | [Live Demo](#)

- Developed a full-stack **MERN** productivity platform featuring dynamic task categorization, interactive calendar integrations, and progress tracking.
- Engineered secure and scalable **REST APIs** with **Node.js** and **Express**, facilitating seamless data flow, robust authentication, and analytics visualization.
- Integrated **Cloudinary** for optimized, cloud-based media storage, significantly reducing server payload while enabling a responsive "Portrait Gallery" for visualizing user goals.

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, C/C++, SQL, HTML/CSS

AI/LLM & GenAI: LangChain, RAG, Agentic AI, Prompt Engineering, OCR, NLP, OpenAI API, Gemini API, Vision-Language Models (VLMs), Multimodal AI, Pydantic, Structured Output Generation

Frontend: React.js, Tailwind CSS, GSAP, JavaScript (ES6+), Responsive UI

Backend: Node.js, Express.js, REST APIs, MongoDB, Authentication, ETL/Data Pipelines

Tools & Platforms: Git, GitHub, Postman, Vercel, Render, Cloudinary, PyMuPDF